AN ABELIAN VARIETY IS COMMUTATIVE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Lemma lemma-abelian-variety-change-field (canonical tag 0BFB). cited at proof lines 1876-1877. Replace k by its algebraic closure.
- Lemma more-morphisms-lemma-flat-proper-family-cannot-collapse-fibre (canonical tag 0AH8). cited at proof lines 1884-1888. Finite fibres over a neighbourhood of e.

Source reference. Original lemma 0BFB: sources/stacks-chow/tags.txt, source lines 14379–14379. Cited at proof lines 1876-1877 to replace k by its algebraic closure; load-bearing for the rational-point argument.

Source reference. Original lemma 0AH8: sources/stacks-chow/tags.txt, source lines 13221–13221. Cited at proof lines 1884-1888 to get the neighbourhood U over which the commutator family is finite; load-bearing.

Lemma 0.2. *An abelian variety is an abelian group scheme, i.e., the group law is commutative.*

Proof. Let k be a field. Let A be an abelian variety over k. By Lemma 0BFB we may replace k by its algebraic closure. Consider the morphism

$$h : A \times_k A \longrightarrow A \times_k A, \quad (x, y) \longmapsto (x, xyx^{-1}y^{-1})$$

This is a morphism over A via the first projection on either side. Let $e \in A(k)$ be the unit. Then we see that $h|_{e \times A}$ is constant with value (e, e). By More on Morphisms, Lemma 0AH8 there exists an open neighbourhood $U \subset A$ of e such that $h|_{U \times A}$ factors through some $Z \subset U \times A$ finite over U. This means that for $x \in U(k)$ the morphism $A \to A$, $y \mapsto xyx^{-1}y^{-1}$ takes finitely many values. Of course this means it is constant with value e. Thus $(x, y) \mapsto xyx^{-1}y^{-1}$ is constant with value e on $U \times A$ which implies that the group law on A is abelian. □

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